# Unit Test Report

Smart Complaint Management System · Team 5D
Software Engineering Mini-Project - Part 3

| | |
|---|---|
| Version | 1.0 |
| Date of execution | 12 Nov 2026, 15:45 |
| Commit tested | 66d1420 |
| Branch | main |
| Test framework | pytest, pytest-cov (coverage.py), pytest-html |
| Overall result | 114 of 114 automated tests passed |

## 1. Summary

96 of 96 unit/module tests passed (100.0%), 0 failed, 0 skipped. Total execution time 177.6 s. Tests are written with pytest and run in isolation: every test gets a fresh temporary data directory, and email/SMS providers are replaced by an in-memory outbox, so no test depends on another or on the network.

## 2. Scope and approach

These are the module-level tests for each feature built in Sprints 1 and 2 (one test file per GitHub issue). They call the service functions directly, or a single API endpoint through Flask's test client, and check one behaviour each. End-to-end flows across modules are reported separately in the System Validation Report. The suite also runs automatically in GitHub Actions on every push and pull request (.github/workflows/ci.yml).

## 3. Results by module

| Module (test file) | Feature / issue | Tests | Passed | Failed | Time (s) |
|---|---|---|---|---|---|
| tests/test_auth.py | #1 Auth & roles | 10 | 10 | 0 | 12.80 |
| tests/test_categorizer.py | #3 AI categorisation | 5 | 5 | 0 | 1.53 |
| tests/test_dashboard.py | #12 Dashboard | 6 | 6 | 0 | 5.17 |
| tests/test_escalation.py | #10 Escalation | 10 | 10 | 0 | 53.97 |
| tests/test_evidence.py | #8 Evidence upload | 8 | 8 | 0 | 29.83 |
| tests/test_notifications.py | #11 Notifications | 12 | 12 | 0 | 29.44 |
| tests/test_resolution.py | #7 Agent resolution | 9 | 9 | 0 | 18.62 |
| tests/test_routing.py | #5 Smart routing | 8 | 8 | 0 | 0.98 |
| tests/test_sla.py | #9 SLA monitoring | 8 | 8 | 0 | 3.62 |
| tests/test_status.py | #6 Status tracking | 7 | 7 | 0 | 14.22 |
| tests/test_submission.py | #2 Complaint submission | 8 | 8 | 0 | 6.21 |
| tests/test_ticket_id.py | #4 Ticket ID | 5 | 5 | 0 | 1.16 |
| Total | | 96 | 96 | 0 | 177.56 |

## 4. Test-by-test results

| # | Test | Result | Time (s) |
|---|---|---|---|
| 1 | test_auth::test_user_can_register | Pass | 0.752 |
| 2 | test_auth::test_password_is_hashed_with_bcrypt | Pass | 0.688 |
| 3 | test_auth::test_duplicate_email_rejected | Pass | 0.702 |
| 4 | test_auth::test_registration_validation | Pass | 0.058 |
| 5 | test_auth::test_self_registration_cannot_pick_admin_role | Pass | 0.745 |
| 6 | test_auth::test_user_can_login_and_gets_jwt | Pass | 1.332 |
| 7 | test_auth::test_wrong_password_rejected | Pass | 1.334 |
| 8 | test_auth::test_jwt_authentication_works | Pass | 1.309 |
| 9 | test_auth::test_role_permissions_enforced | Pass | 3.290 |

| # | Test | Result | Time (s) |
|---|---|---|---|
| 10 | test_auth::test_complaint_list_restricted_to_staff | Pass | 2.590 |
| 11 | test_categorizer::test_top_3_categories_suggested | Pass | 0.071 |
| 12 | test_categorizer::test_confidence_score_present | Pass | 0.007 |
| 13 | test_categorizer::test_correct_category_ranked_first | Pass | 0.007 |
| 14 | test_categorizer::test_user_can_override_suggestion | Pass | 1.373 |
| 15 | test_categorizer::test_empty_text_rejected | Pass | 0.073 |
| 16 | test_dashboard::test_kpis_are_accurate | Pass | 0.001 |
| 17 | test_dashboard::test_filters | Pass | 0.000 |
| 18 | test_dashboard::test_empty_data_has_no_crashes | Pass | 0.000 |
| 19 | test_dashboard::test_dashboard_endpoint_and_permissions | Pass | 2.533 |
| 20 | test_dashboard::test_report_generated_as_csv | Pass | 2.577 |
| 21 | test_dashboard::test_dashboard_page_has_auto_refresh | Pass | 0.055 |
| 22 | test_escalation::test_escalation_moves_ticket_to_priority_queue_with_new_sla | Pass | 5.074 |
| 23 | test_escalation::test_new_sla_timer_raises_fresh_alerts | Pass | 5.167 |
| 24 | test_escalation::test_stakeholders_notified | Pass | 5.039 |
| 25 | test_escalation::test_escalate_to_another_team | Pass | 5.782 |
| 26 | test_escalation::test_escalated_ticket_tops_receiving_agent_queue | Pass | 5.156 |
| 27 | test_escalation::test_validation | Pass | 5.078 |
| 28 | test_escalation::test_only_supervisors_and_admins_escalate | Pass | 6.500 |
| 29 | test_escalation::test_resolved_complaint_cannot_be_escalated | Pass | 5.162 |
| 30 | test_escalation::test_stalled_complaints_listed_for_supervisor | Pass | 5.278 |
| 31 | test_escalation::test_unassigned_complaint_escalated_to_team_gets_assigned | Pass | 5.738 |
| 32 | test_evidence::test_images_and_pdfs_upload_successfully | Pass | 3.970 |
| 33 | test_evidence::test_files_linked_to_complaint | Pass | 3.865 |
| 34 | test_evidence::test_invalid_files_rejected | Pass | 3.937 |
| 35 | test_evidence::test_file_size_validated | Pass | 4.031 |
| 36 | test_evidence::test_only_assigned_agent_or_supervisor_can_upload | Pass | 5.076 |
| 37 | test_evidence::test_download_evidence | Pass | 5.130 |
| 38 | test_evidence::test_closed_complaint_rejects_evidence | Pass | 3.821 |
| 39 | test_evidence::test_file_type_detected_by_signature | Pass | 0.000 |
| 40 | test_notifications::test_submission_notifies_citizen_and_assigned_agent | Pass | 3.904 |
| 41 | test_notifications::test_status_updates_sent_to_citizen | Pass | 3.867 |
| 42 | test_notifications::test_resolution_notification_sent | Pass | 3.908 |
| 43 | test_notifications::test_sms_only_for_opted_in_users | Pass | 3.846 |
| 44 | test_notifications::test_email_opt_out_and_preference_validation | Pass | 3.282 |
| 45 | test_notifications::test_sla_warning_and_breach_notifications | Pass | 2.356 |
| 46 | test_notifications::test_unassigned_complaint_notifies_supervisors | Pass | 1.300 |
| 47 | test_notifications::test_reassigned_agent_notified_on_override | Pass | 2.604 |
| 48 | test_notifications::test_mark_read_and_privacy | Pass | 1.944 |
| 49 | test_notifications::test_provider_failure_does_not_break_request | Pass | 2.419 |
| 50 | test_notifications::test_email_provider | Pass | 0.010 |
| 51 | test_notifications::test_sms_provider | Pass | 0.002 |
| 52 | test_resolution::test_agent_queue_sorted_by_sla_with_overdue_on_top | Pass | 2.278 |
| 53 | test_resolution::test_agent_resolves_with_notes_and_citizen_is_notified | Pass | 2.015 |
| 54 | test_resolution::test_resolving_without_notes_is_rejected | Pass | 2.041 |
| 55 | test_resolution::test_internal_notes_stored_with_author_and_timestamp | Pass | 2.027 |
| 56 | test_resolution::test_internal_notes_hidden_from_citizen | Pass | 1.927 |
| 57 | test_resolution::test_only_assigned_agent_can_manage | Pass | 2.570 |

| # | Test | Result | Time (s) |
|---|---|---|---|
| 58 | test_resolution::test_pending_info_question_reaches_citizen | Pass | 1.901 |
| 59 | test_resolution::test_reopen_clears_resolution_and_done_items_listed_last | Pass | 1.929 |
| 60 | test_resolution::test_closed_complaint_rejects_notes | Pass | 1.935 |
| 61 | test_routing::test_correct_department_for_each_category | Pass | 0.000 |
| 62 | test_routing::test_agent_assignment_is_automatic | Pass | 0.000 |
| 63 | test_routing::test_only_agents_from_the_right_department | Pass | 0.000 |
| 64 | test_routing::test_workload_balancing_picks_least_loaded | Pass | 0.000 |
| 65 | test_routing::test_resolved_complaints_do_not_count_as_load | Pass | 0.000 |
| 66 | test_routing::test_same_zone_agent_preferred | Pass | 0.000 |
| 67 | test_routing::test_unavailable_agents_go_to_supervisor_queue | Pass | 0.001 |
| 68 | test_routing::test_submitted_complaint_is_routed | Pass | 0.983 |
| 69 | test_sla::test_deadline_calculated_from_severity_rules | Pass | 0.001 |
| 70 | test_sla::test_warning_alert_triggers_near_deadline | Pass | 0.001 |
| 71 | test_sla::test_breach_alert_triggers_after_deadline | Pass | 0.001 |
| 72 | test_sla::test_resolved_complaints_do_not_alert | Pass | 0.001 |
| 73 | test_sla::test_violations_and_compliance_tracked | Pass | 0.001 |
| 74 | test_sla::test_new_complaint_gets_sla_deadline | Pass | 0.669 |
| 75 | test_sla::test_admin_configures_rules_others_cannot | Pass | 1.645 |
| 76 | test_sla::test_alert_endpoints | Pass | 1.298 |
| 77 | test_status::test_new_complaint_starts_new_then_assigned | Pass | 1.945 |
| 78 | test_status::test_status_changes_recorded_with_timestamps | Pass | 1.942 |
| 79 | test_status::test_timeline_shows_complete_history | Pass | 1.908 |
| 80 | test_status::test_invalid_transition_rejected | Pass | 1.975 |
| 81 | test_status::test_permissions_on_status_changes | Pass | 2.556 |
| 82 | test_status::test_only_involved_users_can_view | Pass | 3.212 |
| 83 | test_status::test_unassigned_complaint_stays_new | Pass | 0.678 |
| 84 | test_submission::test_complaint_submitted_successfully | Pass | 0.684 |
| 85 | test_submission::test_images_uploaded_correctly | Pass | 0.697 |
| 86 | test_submission::test_gps_coordinates_stored | Pass | 0.807 |
| 87 | test_submission::test_validation_errors_shown | Pass | 0.652 |
| 88 | test_submission::test_oversized_image_rejected | Pass | 0.703 |
| 89 | test_submission::test_non_image_and_too_many_images_rejected | Pass | 0.693 |
| 90 | test_submission::test_login_required_and_only_citizens_submit | Pass | 0.671 |
| 91 | test_submission::test_citizen_sees_only_own_complaints | Pass | 1.307 |
| 92 | test_ticket_id::test_format_is_date_based | Pass | 0.000 |
| 93 | test_ticket_id::test_sequence_increments_within_a_day | Pass | 0.000 |
| 94 | test_ticket_id::test_sequence_restarts_on_new_day | Pass | 0.000 |
| 95 | test_ticket_id::test_no_duplicates_over_many_complaints | Pass | 0.490 |
| 96 | test_ticket_id::test_submitted_complaint_gets_ticket_id | Pass | 0.671 |

## 5. Failures

None.

## 6. Evidence

Raw results are in docs/reports/: junit.xml (machine-readable), unit-test-report.html (pytest-html, open in a browser) and pytest-output.txt (console log). Regenerate with `python scripts/generate_reports.py`.